

1 Summary

Model (Run-name): model1 (test)

Iteration: 30

TAC: 154886.1539978815

Took: 0.931840181350708s

Total Time: 59.62215805053711s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

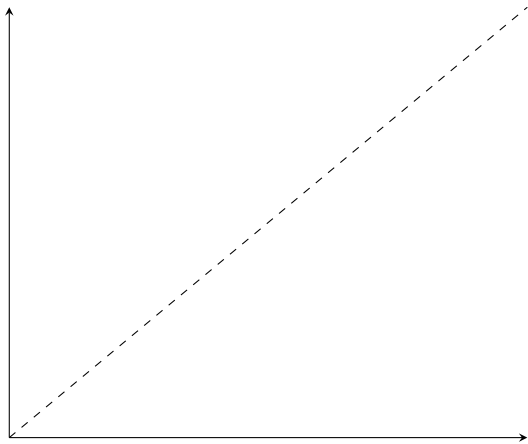
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

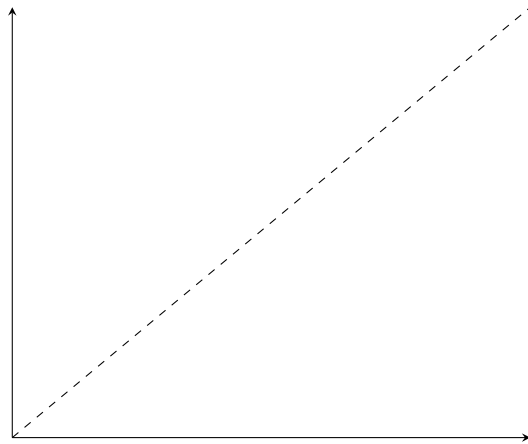
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052800177974459614	$b_{1,1,1}^{H,out}$	2291.976524203233
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017114454331425086	$b_{1,2,2}^{H,out}$	3887.865294196201
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.03076962419187255	$b_{2,1,2}^{H,out}$	9343.202058705698
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014410797366485182	$b_{1,1,1}^{C,in}$	8606.7979412943
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009629211220039808	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022670960232968253	$b_{2,1,2}^{C,in}$	5125.538624086725
$A_{1,1,1}^\beta$	69.49758876855047	$q_{1,1,1}$	662.134705803799	$b_{1,1,1}^{C,out}$	9268.9326470981
$A_{1,2,2}^\beta$	66.74637189255778	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	150.84969099235698	$q_{2,1,2}$	2456.7979412943023	$b_{2,1,2}^{C,out}$	7582.336565381027
$A_{cu,1}^\beta$	5.3386847474643435	$q_{cu,1}$	187.86529419620115		
$A_{cu,2}^\beta$	37.40128299925594	$q_{cu,2}$	1943.2020587056977		
$A_{hu,1}^\beta$	13.064750414488246	$q_{hu,1}$	481.06735290189954		
$A_{1,1,1}$	69.49758876855047	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	66.74637189255778	$t_{1,2}^{(H)}$	583.7865294196201		
$A_{2,1,2}$	150.84969099235698	$t_{1,3}^{(H)}$	388.78652941962014		
$A_{cu,1}$	5.3386847474643435	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.40128299925594	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.064750414488246	$t_{2,3}^{(H)}$	467.1601029352849		
$dt_{1,1,1}$	32.071156860126635	$t_{1,1}^{(C)}$	617.9288431398734		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	573.7865294196201		
$dt_{1,2,2}$	83.78652941962014	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	38.78652941962011	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	16.213470580379862	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	57.1601029352849	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	68.78652941962011	$t_{1,1,1}^H$	504.30895275685174		
$dt_{cu,2}$	147.1601029352849	$t_{1,2,2}^H$	388.7865294196201		
$dt_{hu,1}$	62.071156860126635	$t_{2,1,2}^H$	467.1601029352849		
$f_{1,1,1}^H$	4.544786507703126	$t_{1,1,1}^C$	617.9288431398733		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	606.523181231542		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	2954.1112300070317		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5837.865294196201		
$f_{2,1,2}^C$	12.501313717284695	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

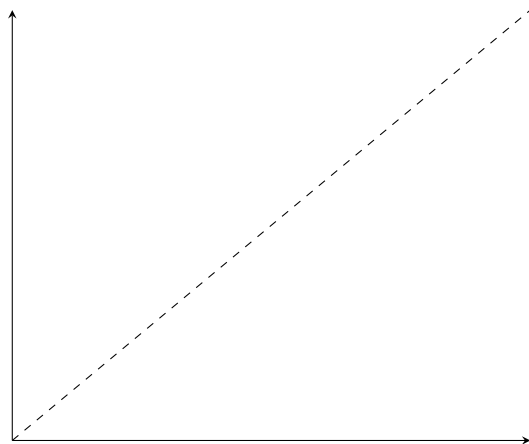
- $(1, 1, 1)$
 - Value: 0.052800
 - Error: 0.000000497604766
 - Relative error: 0.000009424212101
- $(1, 2, 2)$
 - Value: 0.017114
 - Error: 0.000001084365079
 - Relative error: 0.000063355591555
- $(2, 1, 2)$
 - Value: 0.030770
 - Error: 0.000002472740040
 - Relative error: 0.000080356566074
- $cu, 1$
 - Value: 0.014411
 - Error: 0.000000190326414
 - Relative error: 0.000013207034697
- $cu, 2$
 - Value: 0.009629
 - Error: 0.000000453067589
 - Relative error: 0.000047049157202
- $hu, 1$
 - Value: 0.022671
 - Error: 0.000000003578914
 - Relative error: 0.000000157863334



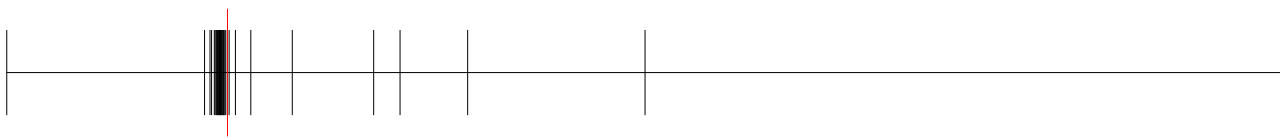
(a) *RecLMTD* - (1, 1, 1)



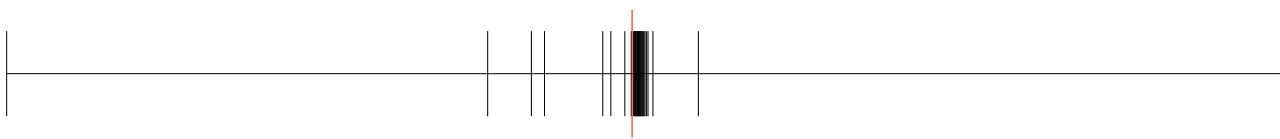
(b) *RecLMTD* - (1, 2, 2)



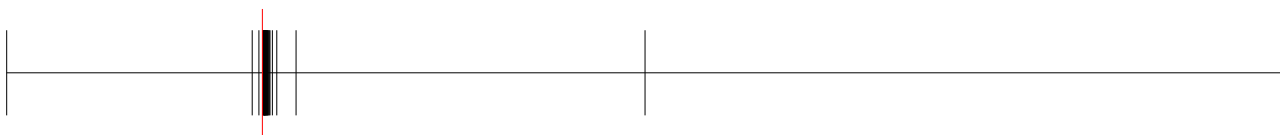
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 2954.1112300070317
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5837.865294196201
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 2291.976524203233
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3887.865294196201
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9343.202058705698
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8606.7979412943
 - Absolute error: 1.8189894035458565e-12
 - Relative error: -2.1134333766784287e-16
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 5125.538624086725
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9268.9326470981
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7582.336565381027
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.1994913361211687e-16



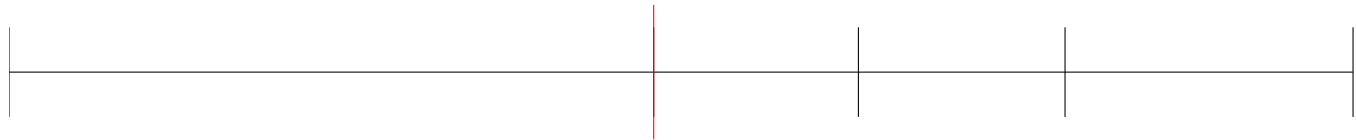
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



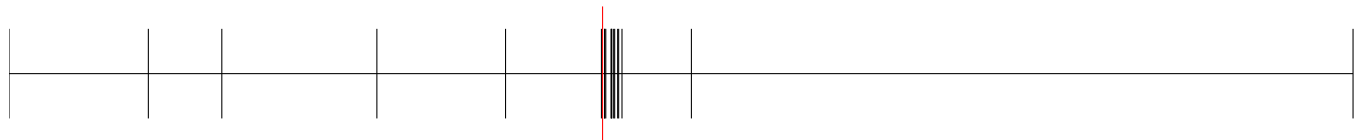
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



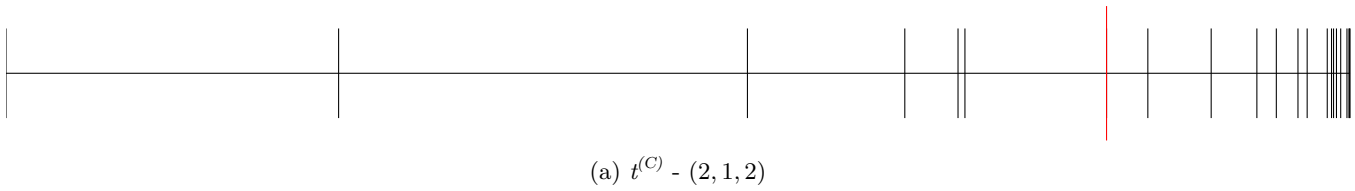
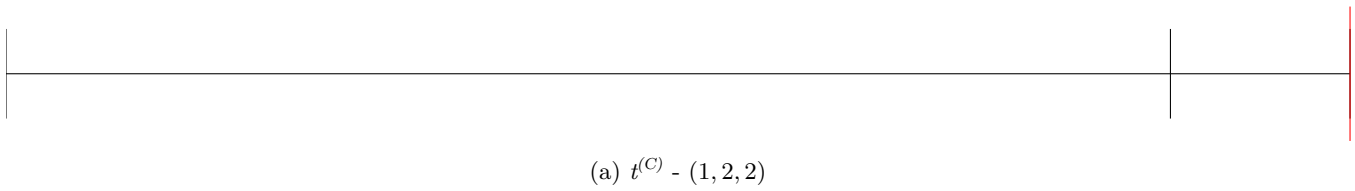
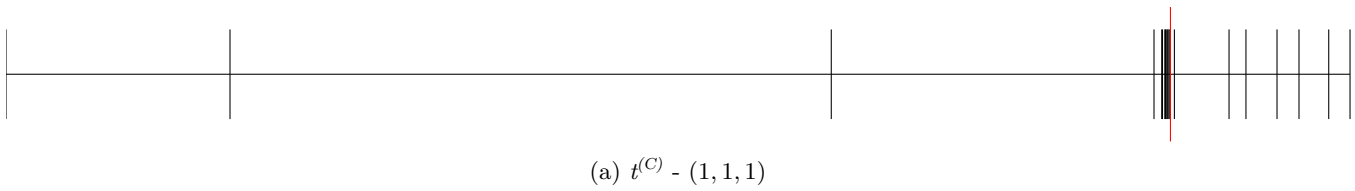
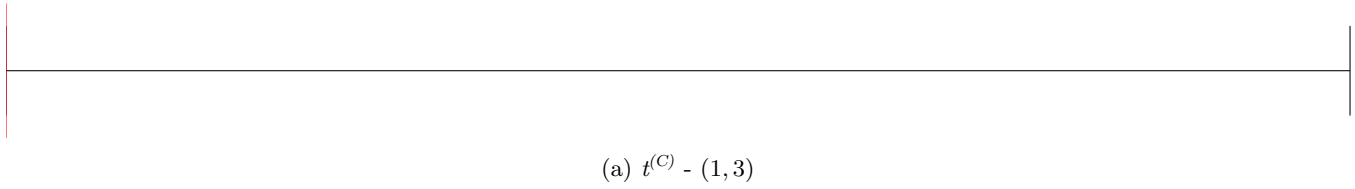
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

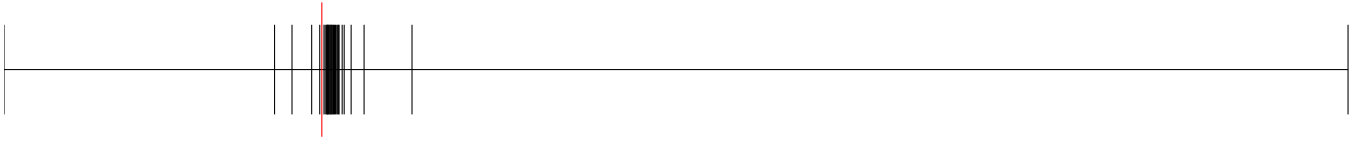
- $(1, 1, 1)$
 - Value: 69.497589
 - Error: 0.424071850463619
 - Relative error: 0.006064956791777
- $(1, 2, 2)$
 - Value: 66.746372
 - Error: 0.000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 150.849691
 - Error: 0.339807745626700
 - Relative error: 0.002247561824486

5.2 A_{CU}

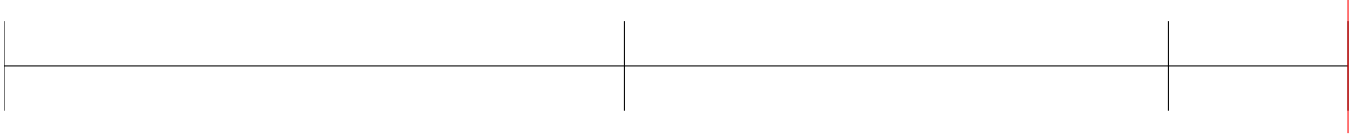
- $cu, 1$
 - Value: 5.338685
 - Error: 0.075892626248815
 - Relative error: 0.014016352710603
- $cu, 2$
 - Value: 37.401283
 - Error: 0.021723133730774
 - Relative error: 0.000580475380668

5.3 A_{HU}

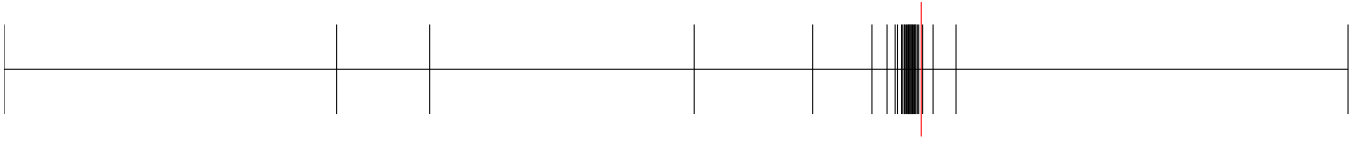
- $hu, 1$
 - Value: 13.064750
 - Error: 0.022760177933675
 - Relative error: 0.001739076180527



(a) $A^\beta - (1, 1, 1)$



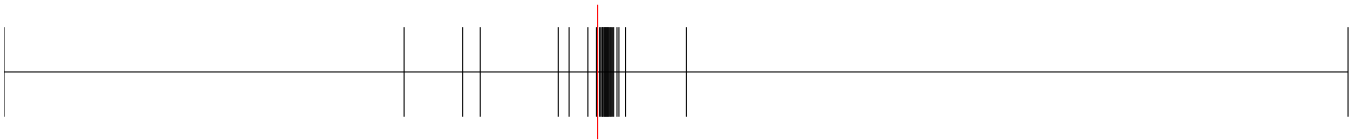
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$



(a) $A^\beta - hu, 1$

6 A^β Breakpoints

- (1, 1, 1)
 - q : 69.497589
 - $\hat{q}^\beta$: 69.497589
 - q^β : 69.497589
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, **69.42777867315647**, **69.89902342903967**, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 66.746372
 - $\hat{q}^\beta$: 66.746372
 - q^β : 66.746372
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, **66.17325582523148**, **67.27673099173637**, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 150.849691
 - $\hat{q}^\beta$: 150.849691
 - q^β : 150.849691
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, **149.84689443310864**, **720.0**]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 5.338685
 - $\hat{q}^\beta$: 5.338685
 - q^β : 5.338685
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335,

4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.0892956597199035, **5.195781532076625**, **7.702824927089129**, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $cu, 2$

– q : 37.401283

– $\hat{q}^\beta$: 37.401283

– q^β : 37.401283

– Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, **37.21616748032837**, **37.443498367229196**, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $hu, 1$

– q : 13.064750

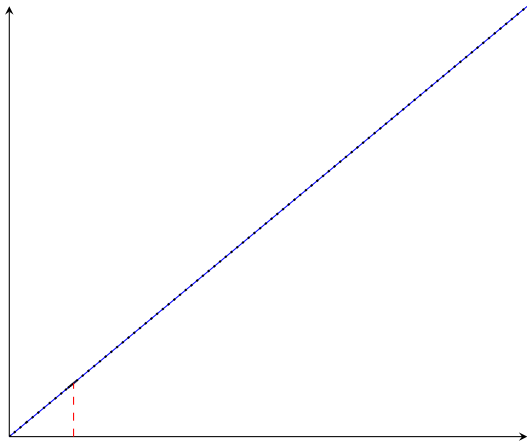
– $\hat{q}^\beta$: 13.064750

– q^β : 13.064750

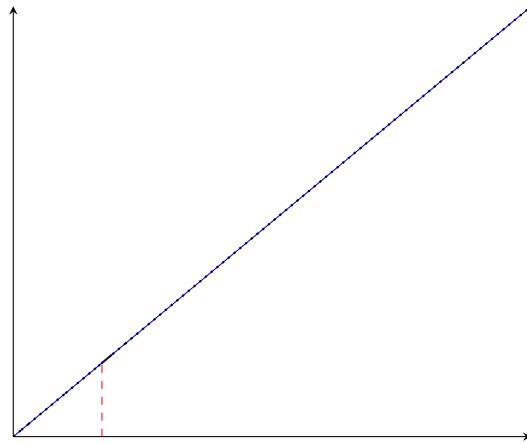
– Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, **12.917705220652834**, **13.104034300095861**, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]

– Error: 0.0000000000000000

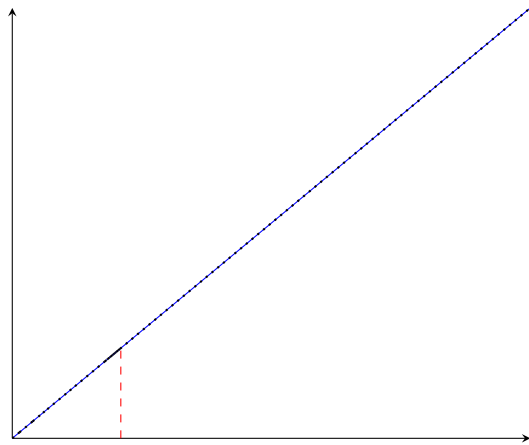
– Relative error: 0.0000000000000000



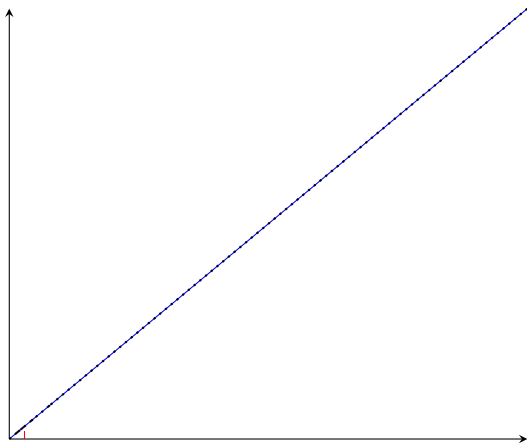
(a) $q^\beta - (1, 1, 1)$



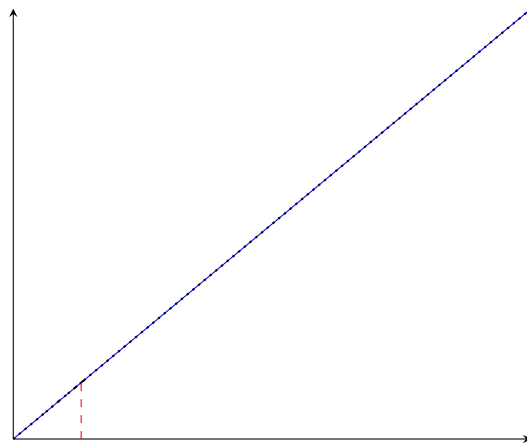
(b) $q^\beta - (1, 2, 2)$



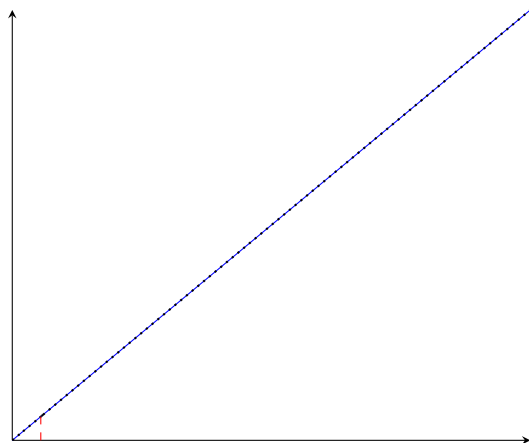
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1024.4613759132749
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.0080377413232129	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.004340277777777778	$b_{1,1,2}^{C,out}$	1024.4613759132753
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	564.6191576569319		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	576.9059677174985		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	502.18150166658575		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	466.6292188464114		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	3545.8887699929683		
$f_{1,2,1}^H$	5.455213492296874	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	2.4986862827153047	$b_{1,2,1}^{H,out}$	3545.888769992968		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		